

PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	Solution Architecture	DATE	July 2026

Solution Architecture

Architecture Overview

The system follows a three-tier architecture: a data/model layer, an application (backend) layer, and a presentation (frontend) layer.

1. Data & Model Layer

- Raw data: application_record.csv, credit_record.csv
- Preprocessing pipeline: cleaning, merging, encoding, feature engineering (Google Colab)
- Trained model artifact: model.pkl (Random Forest Classifier)

2. Application Layer (Backend)

- Flask application (app.py) handles routing and request processing
- Loads model.pkl using Pickle at application startup
- Encodes incoming form data to match the model's trained feature format
- Calls model.predict() and returns the result to the frontend

3. Presentation Layer (Frontend)

- home.html: landing page with project overview
- index.html: input form for applicant details
- result.html: displays the prediction outcome

Architecture Diagram (Textual)

[User Browser] <--> [Flask App (app.py)] <--> [model.pkl (Random Forest)]

Flask renders home.html, index.html, and result.html based on the route and request method (GET/POST).